DISTANCE WITHOUT FLATNESS: METRIC-DRIVEN CLUSTERING ON SPHERES, TORI AND HYPERBOLIC SPACES

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The K-means algorithm [1] minimizes the within cluster sum of squared Euclidean distances, and its centroid update relies on the linear structure of $\mathbb{R}^n$. On curved surfaces this assumption breaks and squared Euclidean cost no longer reflects the intrinsic separation between points. Here we revisit the algorithm under three non Euclidean geometries and study how the choice of metric reshapes the resulting clusters. On the sphere S^2 we use the Haversine distance [2], on the flat torus $\mathbb{T}^2$ the periodic Euclidean metric obtained by reducing coordinate differences modulo 2π along each angular direction [3], and on the hyperbolic Poincaré disk $\mathbb{D}$ we use the Poincaré metric induced by the conformal factor $2/(1-\|x\|^2)$ on the open unit disk [4].

We view this work as a first step toward systematic clustering on non Euclidean spaces. The three surfaces studied here represent positive curvature, flat periodicity, and negative curvature in their simplest forms, echoing the role that curvature plays in shaping geometric analysis on Riemannian configuration spaces more broadly [5], and the framework extends naturally to product manifolds, Stiefel and Grassmann manifolds, spaces of symmetric positive definite matrices, and further Riemannian settings encountered in modern data analysis [6].

References

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